Sanyam Sharma

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Work Experience

Indian Institute of Technology, Hyderabad — Hyderabad, Telangana
Research Intern — Jan 2026 – Present
- Developed an AI-driven network automation system translating natural language intents into SONiC configurations
- Integrated Batfish verification and GNS3 digital-twin testing for automated network change validation
- Built GNS3 REST API-driven workflows for automated network topology discovery and configuration analysis in SONiC environments
- Created Terraform code-generation datasets and benchmarked SLMs (small language models) using CodeBLEU, CodeBERTScore and Pass@k metrics

Indian Institute of Technology, Mandi — Mandi, Himachal Pradesh
Summer Intern — Jun 2025 – Jul 2025
- Generated speech embeddings using MFCC, wav2vec2.0 and wav2vec2-phoneme for language identification
- Implemented a Conformer-based PyTorch architecture achieving 79% accuracy on unseen multilingual speech samples
- Developed a PyQt6 desktop GUI integrating pretrained AASIST and RawNet models for real-time spoofed speech detection and inference

Indian Institute of Technology, Mandi — Mandi, Himachal Pradesh
Research Intern — Jun 2024 – Jul 2024
- Set up data pipeline using Telegraf and InfluxDB to collect and store sensor data from an ESP32 via MQTT
- Deployed the system on a Raspberry Pi using Docker for real-time data processing

Education

Jawaharlal Nehru Government Engineering College — Mandi, Himachal Pradesh
B.Tech – CSE (AI & ML) — Aug 2022 – May 2026

Government Senior Secondary School, Kangoo — Hamirpur, Himachal Pradesh
Senior Secondary (Non-Medical) — Mar 2022

Projects

Llama-3.2-1B Text-to-SQL | **HuggingFace** | **Blog** — 2026
Tech: Python, PyTorch, Unsloth, PEFT, QLoRA, Hugging Face
- QLoRA fine-tuned Llama-3.2-1B-Instruct (4-bit NF4, LoRA r=16) via SFT on 73K text-to-SQL examples; exact-match accuracy increased from 40.1% to 84.3% - a 2.1x improvement over the untuned baseline
- Published LoRA adapter and GGUF-quantized variant to Hugging Face Hub with usage examples and evaluation results

Transformer Serve | **LiveDemo** | **GitHub** | **Docker** — 2026
Tech: Python, FastAPI, Docker, PyTorch, Hugging Face Transformers
- Built a model-agnostic inference API serving any Hugging Face causal or seq2seq text-generation model
- Containerized with Docker (CPU-only PyTorch build) and deployed a public API on Hugging Face Spaces, with 200+ Docker Hub pulls

AI-Powered Assistive Vision for Blind People - Major Project — Aug 2025 – Dec 2025
Tech: gTTS, FaceNet, Gemini-API, pytesseract, Raspberry Pi
- Developed an AI-powered assistive vision smart cap for blind individuals using Raspberry Pi, camera, LiDAR, audio module
- Implemented OCR-based text recognition to read printed text and deliver real-time audio output
- Integrated obstacle detection with audio-based navigation for safe mobility

Extension for malicious QR code and URL detection - Capstone Project — Jan 2025 – May 2025
Tech: Python, Flask, Scikit-learn, JavaScript
- Engineered features from consolidated dataset of 14,013 URL/QR samples across malicious and legitimate classes
- Built a stacked ensemble ML model (5 base + meta-classifier), achieving 91.3% accuracy and 95.15% precision
- Developed a Chrome extension with Flask backend for real-time malicious URL and QR code detection

Technical Skills

Languages: Python, C++, SQL, HTML, CSS
Libraries: PyTorch, Transformers, Hugging Face, Unsloth, PEFT, TRL, FastAPI, Flask, Scikit-learn, NumPy, Pandas
Tools: Docker, llama.cpp, Weights & Biases, Git, GitHub, Linux, Postman, Ollama, Ansible, SONiC, GNS3, Batfish

Publications

HybridStack-MLP: Advanced Ensemble Learning for Malicious QR Code and URL Detection — 2025
- *Proc. 2nd IEEE International Conference for Women in Computing (InCoWoCo 2025)*, IEEE Xplore, 2025. doi:10.1109/InCoWoCo68239.2025.11407094

Awards

Winner - State Level Smart Hackathon (SLSH), Himachal Pradesh — 2025